Relationship Between Coronal Mass Ejection Observables Near the Sun and Measurements Near Earth

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INTRODUCTION

As CMEs travel from the Sun to Earth, they are altered by solar wind, magnetic fields, and shocks, making early predictions difficult. This study examines how CME speed, direction, and geometry change during propagation, aiming to bridge the gap between solar observations and Earth impacts by analyzing events across multiple solar cycles.

OBJECTIVES

Aim
Investigate links between near-Sun and Earth CME parameters for improved impact prediction.
Objectives

- Link speed, acceleration, and initial velocity.
- Assess compression, deflection, and drag effects.
- Correlate near-Sun properties with Dst index.
- Analyze source location impact on direction and geoeffectiveness.
- Use standardized multi-source datasets (e.g., GCS, JHELIOVIEWER).

METHODOLOGY

Analyzed 292 CME–ICME pairs from Solar Cycles 23–25, selected for clear Sun–Earth connections using strict criteria.

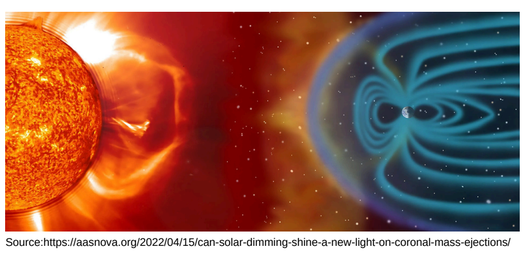
Near-Sun Data: Extracted from SOHO, STEREO, JHelioviewer, CDAW, and GCS modelling (where available).

Near-Earth Data: Collected from NASA OMNIWeb, Richardson & Cane catalogue, and GMU CME Scoreboard.

Each event was standardized with validation, time alignment, imputation, unit unification, and coordinate conversion.

Tools Applied :

Correlation (**Pearson**), regression (**Gaussian fits for Dst**), cycle comparisons, source mapping, and interactive outlier detection (**Plotly**) were performed in **Python** using open-source tools in **Jupyter**.

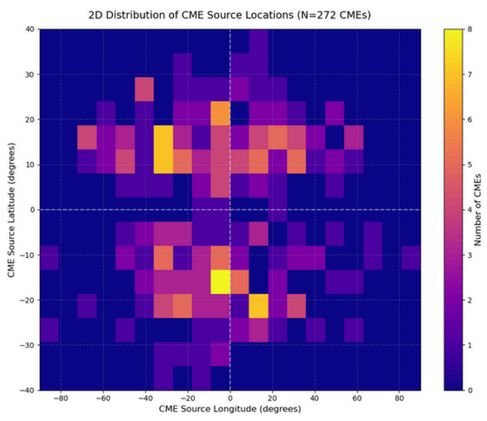


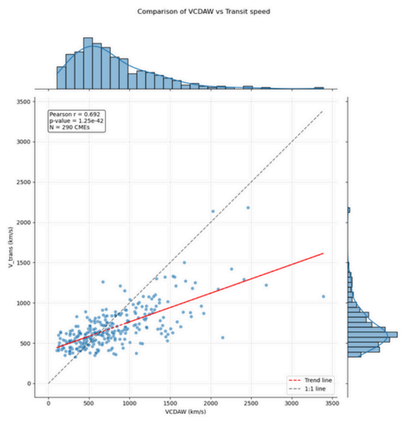
Source:https://aasnova.org/2022/04/15/can-solar-dimming-shine-a-new-light-on-coronal-mass-ejections/

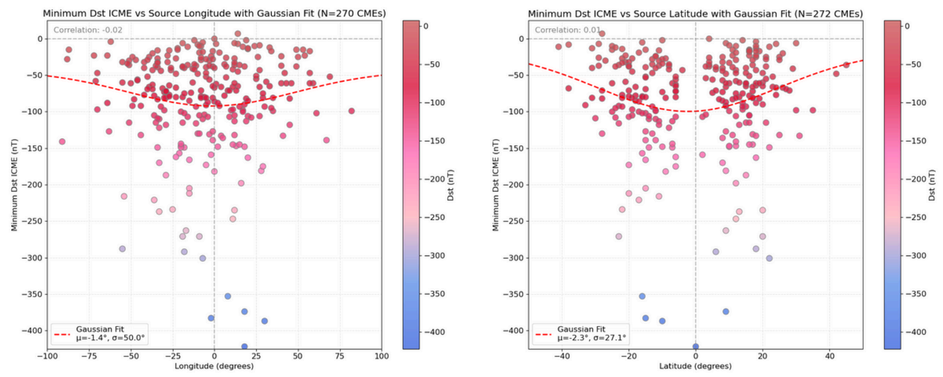
RESULTS

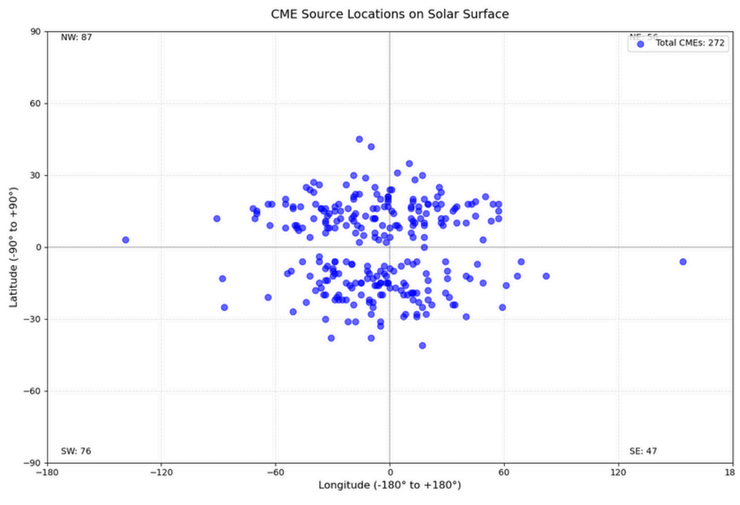
What do solar eruption traits tell us about Earth impacts?

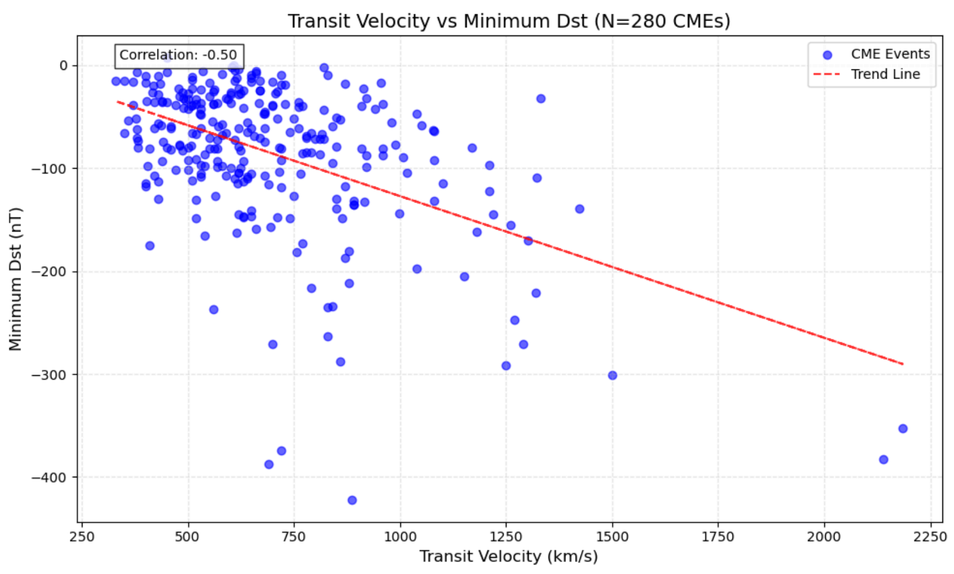
- **Source Locations** - Most CMEs originated within ±30° latitude. Western hemisphere events showed stronger geoeffectiveness.
- **CME Speed vs. Transit Speed** - Strong correlation (r = 0.69); faster CMEs near the Sun tend to arrive faster at Earth.
- **Transit Speed vs. Dst (Storm Strength)** - Moderate negative correlation (r = -0.50); higher speeds linked to stronger geomagnetic storms.
- **Source Coordinates vs. Geoeffectiveness** - Latitude and longitude showed weak correlations, but Dst distributions skewed toward equatorial and western sources.











All findings are supported through regression analysis cycle-based comparisons and interactive visualizations.



Source:https://aasnova.org/2022/04/15/can-solar-dimming-shine-a-new-light-on-coronal-mass-ejections/

DISCUSSION

The results reinforce that CME initial speed is a strong predictor of both transit velocity and geomagnetic storm intensity. While source coordinates showed weak direct correlations with storm strength, most impactful CMEs originated within ±20° latitude and ±30° longitude—regions magnetically connected to Earth and often associated with active regions.

The slightly higher geoeffectiveness of western hemisphere CMEs may be explained by solar magnetic field line orientation, which favors Earth-directed propagation from these regions. These findings support the continued use of near-Sun kinematic and positional data in forecasting models and highlight the importance of monitoring eruptions near the solar disk center.

CONCLUSION

Fast CMEs are more likely to trigger strong geomagnetic storms. Early solar observations—especially speed and direction—can offer reliable clues about space weather impacts.
Bridging the observational gap from Sun to Earth is essential for building faster, smarter forecasting systems.

FUTURE WORK

- Apply machine learning to improve prediction accuracy.
- Incorporate real-time data from Solar Orbiter & Parker Solar Probe.
- Extend analysis to Solar Cycle 25 as more data becomes available.
- Explore deeper integration of GCS parameters for 3D structural modelling.

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